YouTube Comments Engagement & Sentiment Analysis

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Key Results

- Analyzed 4,148 YouTube comments from 29 channels and 49 videos.
- Median comment likes and replies were both zero, showing that most comments received little engagement.
- Positive sentiment was the most frequent classification, appearing in 1,737 comments.
- A scaling experiment showed that Pandas successfully processed up to 2 million rows in Google Colab.

1. Dataset and Provenance

The dataset contains 4,148 YouTube comments and 15 columns, including comment text, likes, replies, timestamps, video titles, channels, content types, languages, engagement and sentiment.

It covers 29 channels, 49 videos, 6 content types and 3 languages. Each record represents one YouTube comment. The source, license and collection method were not documented in the available dataset information. Therefore, the findings describe this dataset only and should not be generalized to all YouTube users or channels.

2. Data Checks and Cleaning

The original dataset contained 4,148 rows and 15 columns, with no missing values. The cleaning process included converting numeric columns, checking negative values, parsing timestamps, standardizing categorical values and checking for duplicate records. The cleaned dataset and cleaning log were saved in the submission folder.

3. Summary Statistics

Metric	Mean	Median	Maximum
Comment likes	14.17	0	21,108
Comment replies	0.60	0	248
Engagement	14.77	0	21,185

The results show a highly uneven distribution of engagement: most comments receive no likes or replies, while a small number receive exceptionally high engagement.

Sentiment classification identified 1,737 positive comments, making positive sentiment the most common category.

4. Findings

Finding 1: Comment engagement is highly concentrated.

The median likes and replies are both zero, while the maximum likes value is 21,108. This indicates that a small number of comments receive much more attention than the majority.

Finding 2: Positive sentiment is the most common classification.

Out of 4,148 comments, 1,737 were classified as positive. This suggests that positive reactions are common in the collected comments, although automated sentiment labels may not always capture context accurately.

Finding 3: Engagement can be compared across channels and content types.

The dataset includes 29 channels and 6 content types. Comparing their likes, replies and engagement can help identify patterns in audience interaction.

5. Marketing Recommendation

Content creators and marketing teams can use these findings to investigate which video topics and channels generate stronger audience interaction. They should consider both engagement and sentiment when planning future content. Additional information, such as video views, impressions and audience size, would allow fairer comparisons between channels and videos.

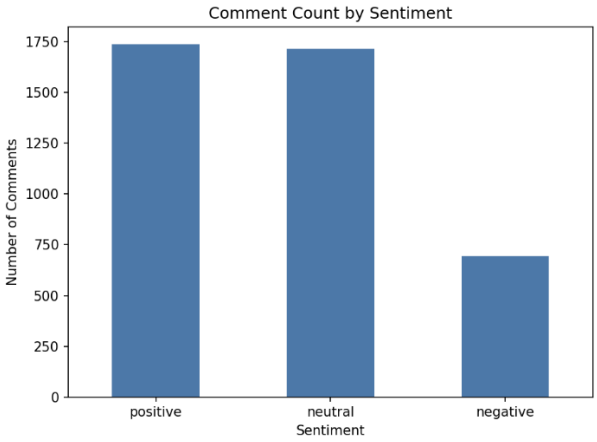
6. Limitations

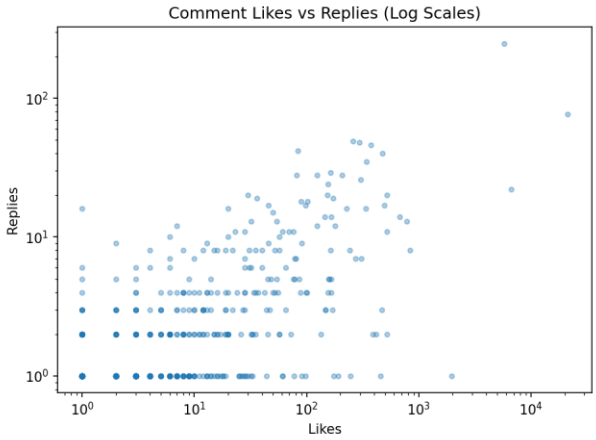
The dataset does not include video views, impressions or channel audience size, so a true engagement rate cannot be calculated. The source, license and collection method are also undocumented. Sentiment labels are model-generated and may contain errors. Since the analysis is observational, it cannot establish that a particular channel or content type causes higher engagement.

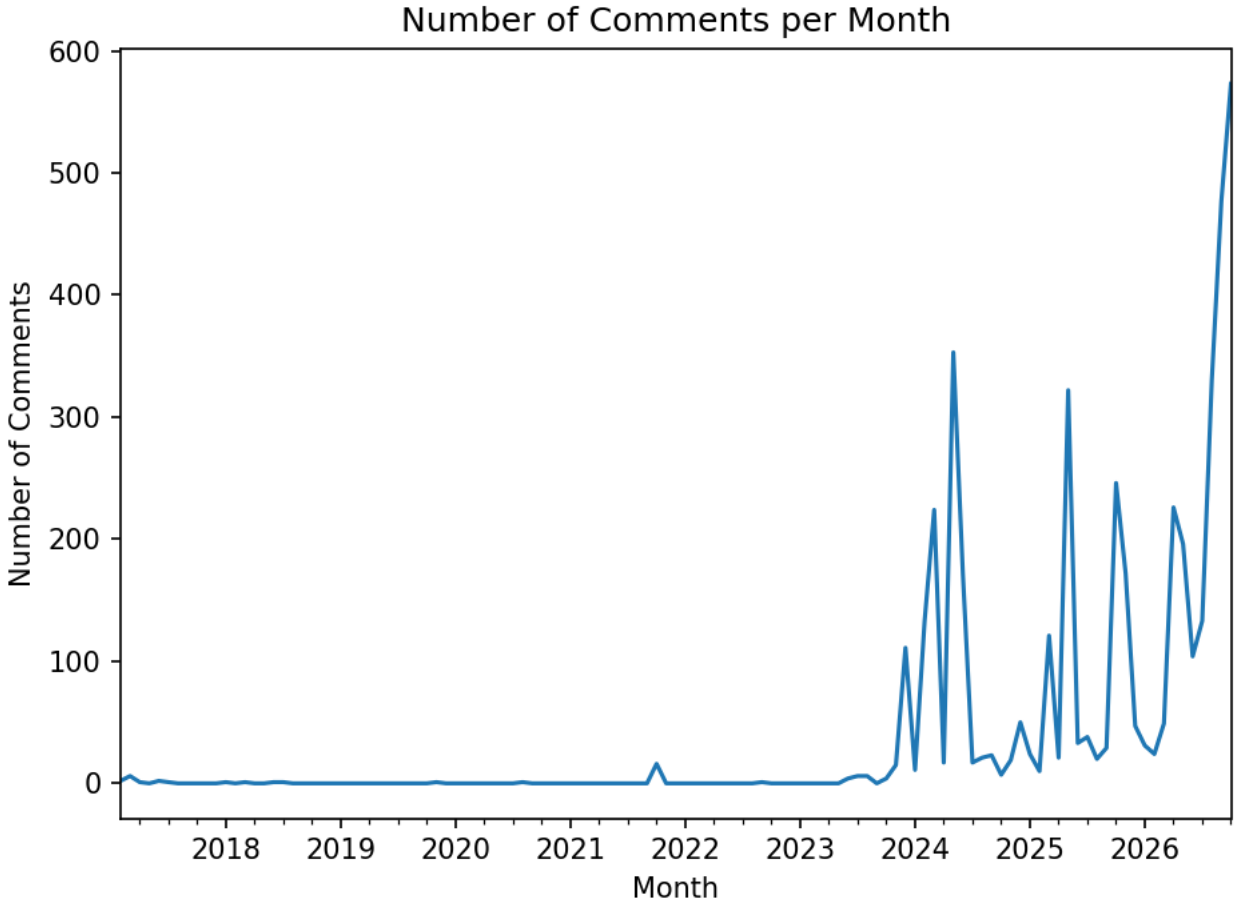
7. Scaling Experiment

The original dataset was repeated to simulate datasets of 2,000, 20,000, 200,000 and 2,000,000 rows. This was a technical experiment, not additional real-world data.

At 2 million rows, the CSV file was 436.2 MB, DataFrame memory usage was approximately 1.30 GB, loading took 8.242 seconds and analysis took 0.379 seconds. Pandas handled all tested sizes successfully in Google Colab. Spark may be worth investigating if future datasets exceed the memory or processing capacity of a single machine.


Comment Count by Sentiment
Number of Comments
1750
1500
1250
1000
750
500
250
0
positive
neutral
negative
Sentiment


Comment Likes vs Replies (Log Scales)
Replies
10^2
10^1
10^0
10^0
10^1
10^2
10^3
10^4
Likes


Number of Comments per Month
Number of Comments
600
500
400
300
200
100
0
2018
2019
2020
2021
2022
2023
2024
2025
2026
Month